Electrical and Computer Engineering

2025-2026

Important Dates for Senior Independent Work (ECE 499)

- Online ECE Funding Proposal Forms due **Friday, February 13, 2026: ECE Funding Application**. You must submit a SEAS funding application to obtain funding in ECE (See separate email from Traci Miller regarding SEAS funding)
- Send poster for printing **by noon on Thursday, April 30, 2026 (Directions forthcoming)**
- ECE 499 (Spring) Poster presentation (10% of grade for ECE 499) will be held on **Monday, May 4, 2026 at 1:30 - 3:00 pm in the Friend Center Convocation Room 113 and upper atrium**
- ECE 499 (Spring) (90% of grade) is due as a PDF on **Monday, April 13, 2026** to adviser, second reader (if applicable) by email. Also submit to UG office here: Submission Form
- **Final submission carries automatic penalty of -⅓ letter grade if submitted late, plus an additional -⅓ letter grade if submitted more than one week late**

Important Dates for Spring Independent Work (ECE 299 and ECE 399)

- Online Project Proposal Form is due **Wednesday, February 11, 2026: Project Proposal Form**
- Online ECE Funding Proposal Forms due **Friday, February 13, 2026: ECE Funding Application**. Juniors must submit a SEAS funding application to obtain funding in ECE (See separate email from Traci Miller regarding SEAS funding)
- Send poster for printing **by noon on April 27, 2026 (Directions forthcoming)**
- Poster presentation (10% of grade) will be **Wednesday, April 29th, 12:00 to 1:30 pm in the E-Quad, 3rd floor B and C wings**
- Spring IW report (90% of grade) is due on **Tuesday, April 28th by 11:59** pm to your adviser, second reader (if applicable) by email. Also, submit to UG office here: Submission Form

You are responsible for reading the ECE IW/Thesis Handbook to make sure you understand all aspects of the course(s) and to check Canvas regularly for important updates

- Is IW mandatory? **Senior thesis (ECE 498/499) is mandatory.** Sophomores and juniors are not required to take IW.
- How do I register for IW? Seniors register for ECE 498 in fall and ECE 499 in spring. Sophomores register for ECE 298 in fall and ECE 299 in spring. Juniors register for ECE 398 in fall and ECE 399 in spring.
- Are the poster sessions mandatory? Yes, each student must present their research at the appropriate time each semester (see above dates)
- What if I can't find an IW/Senior thesis advisor? The following website is a good resource of projects that ECE faculty are interested in advising: https://ece.princeton.edu/academics/undergraduate/student-projects/independent-project-ideas

  If you are having difficulty, contact either the Undergraduate Program Coordinator (Lori Bailey), IW/Senior Thesis Coordinator (Jean Bausmith) faculty instructor for the Independent Work ECE courses (Paul Prucnal) or Directors of Undergraduate Studies (Prateek Mittal, Alejandro Rodriguez, Hossein Valavi)

- Do seniors need a second reader for their thesis? Seniors only need a second reader if their primary advisor is outside of the ECE department. You will need to secure an ECE faculty member to serve as a second reader.
- Do sophomores and juniors need a second reader? No, second readers are only needed if the primary adviser is outside of the ECE department. You will need to secure an ECE faculty member to serve as a second reader.
- How do I print my poster? Directions will be sent out as to how to get the poster printed and the timeline to do so
- Does my senior thesis need to be bound? No, this is not a department requirement
- Can I take IW/Thesis as a P/D/F? Independent work cannot be taken P/D/F. It cannot be used to fulfill the ECE breadth or concentration requirements, but it can be counted as a Department course for juniors and seniors.

- Can IW/Thesis count towards my design requirement? A design-oriented project can satisfy the ECE design requirement. Your project adviser will determine whether or not your particular project satisfies this requirement. See handbook for further details.
- Can I get funding for my IW/Thesis project? Yes, limited funding is available through SEAS and the ECE department. **The deadline is Friday, February 13, 2026** and the form is **ECE Funding Application**. Juniors and Seniors must apply to SEAS for funding as well (see email from Traci Miller regarding SEAS funding)
- Can my research involve human and/or animal subjects? Yes, if you meet certain criteria. Please refer to page 7 of the IW/Thesis handbook.
- What should my interim or final report look like? Guidelines are in the handbook (pages 13 - 17) Also, see page 7 for Engineering Standards
- I can't reach my advisor. What should I do? Not hearing back from an adviser is not an excuse to delay your work. Try emailing, dropping by their office or asking for an appointment from their faculty assistant. View the faculty assistants on our ECE website
- How do I get updated information on IW/Senior thesis? You are responsible for checking Canvas for updates to any information in the handbook including deadlines and submissions. Check Canvas regularly.
- What do I do if I need an extension? Extensions are not normally granted. See page 16 of the handbook for further information. **If no extension is formally granted and you are late in handing in your final report your grade will be adjusted by -⅓ letter grade. For example, a report that is otherwise graded as an A will be reduced to an A-. Reports that are more than one week late will suffer an additional -⅓ letter grade penalty.**
- Are there TAs to help if I get stuck in my research? In order to keep students on track, each **senior** will need to check-in with TAs. There will be four two-week periods for check-ins, and students must check in **at least twice**. Feb 2 - Feb 13, Feb 16 - 27, Mar 2 - 20 (Inc Spring Break), Mar 23 - Apr 3. **TA check-ins cannot be rescheduled! Failure to check-in with TAs regularly will result in -1/3 letter grade for missing more than 3/6 TA check-ins without advance notice.**
- Are there penalties if I don't hand in forms on time or fail to meet with the TAs? Yes, because on-time submission of proposals and regular meetings with instructional staff are mandatory, grade penalties will be applied to students who do not meet departmental deadlines and expectations. **Failure to hand in Project Proposal on time will result in an automatic penalty of -0.1/4.0 of your final grade if submitted late, plus an additional -0.1/4.0 for every additional week late.**

INTRODUCTION

Electrical and computer engineers are well positioned to address a variety of the crucial engineering issues facing societies today. Think, for example, about:

Energy: efficient lighting, buildings, computing, and communications; solar cells; low power devices and circuits; energy harvesting from the environment; smart power grids;
Information: security; communications; networks; machine learning;
Health: informatics; medical imaging, sensing, and diagnosis; brain imaging;

Princeton's Department of Electrical and Computer Engineering is actively engaged in teaching and research on many topics related to these challenges.

Junior and Senior independent work is an experimental, computational and/or theoretical study of an important problem that is investigated by an individual student or team of students outside of a structured lecture course. It is open ended, work is not circumscribed by syllabus, text, problem sets, etc. It can also have one or several different components (experiment, theory, computation, engineering design, to name but a few).

Independent work is often extremely challenging on both a personal and academic level, but is also very fulfilling. Because it is not circumscribed, more is expected from you. On the other hand, most students consider their independent work experience – working with one or more faculty members on a challenging problem – to be one of the high points of their Princeton education in providing a greater sense of accomplishment. The objectives of this guide are to:

1. give a brief overview of Electrical and Computer Engineering independent work.
2. explain the graded elements of independent work and certain rules and procedures governing the preparation of the final report or thesis.
3. help develop a schedule that will avoid the typical rush that occurs in completing independent work at the end of the semester.
4. provide some tips on how to organize your research and then write up your results in your final report.

Selecting a Project Topic, finding an adviser, and Working with a Faculty Member

Students seek out independent work topics based on their interest in research within the department. The student then proceeds to discuss possible project topics with the faculty member supervising the research. Students may also suggest their own topics. If a student is not able to obtain a firm commitment from a faculty adviser, the Director of Undergraduate Studies, with the assistance of the faculty, will then make the final assignment of students to faculty, trying to satisfy student requests and yet maintain a reasonable distribution of students among all faculty members. The faculty adviser can be from outside the ECE department. The primary adviser can also be from outside the University (e.g. a local company). In that case a secondary-adviser has to be selected from among the Electrical and Computer Engineering faculty. The following website is a good resource of projects that ECE faculty are interested in advising:
https://ece.princeton.edu/academics/undergraduate/student-projects/independent-project-ideas

When working on senior thesis or independent work, it is recommended that the student is proactive about meeting with their adviser and keeping their adviser aware of successes, challenges, and progress throughout the semester. Likewise, it is recommended that students are proactive about responding to feedback from their adviser.

Enrolling in Independent Work

To enroll in one semester of Independent work you must:

1. Register for ECE 298 or 398 (fall) or ECE 299 or 399 (spring)
2. Seniors must register for 498 (fall) and 499 (spring)
3. Prior to the start of the semester, reach an agreement with a faculty adviser on a project and have this information recorded in the Undergraduate Program office.

The independent work timetable will be distributed to students at the beginning of the semester.

Senior Independent Work Funds
Research and Research Related Conferences

Limited funding is provided by both the Electrical and Computer Engineering Department and the School of Engineering for independent work requiring financial support for acquisition of data, or other special requirements. Awards are typically modest and not all proposals can be funded. Talk with your adviser if you feel you may qualify for this support. Remember, however, that the application deadline is relatively early, implying that you must already have a good idea of what you need and why it requires support. If you would like to apply, you must submit a one or two page proposal, including a budget, to your adviser who must then sign off on the funding application and forward it on to the Director of Undergraduate Studies. **ECE Funding Application**

The Electrical and Computer Engineering Department also has limited senior thesis/independent work funding "Excellence in Engineering" which is available for both research and conference attendance. This funding encourages student attendance at electrical and computer engineering conferences located in both the USA and Canada. *The selected conference must have components of professional development such as tutorials, plenary talks, poster presentation, etc*. The application and a waiver form for this funding can be requested to the Undergraduate Coordinator's office.

Satisfying Requirements

Design Requirement

A design-oriented project can satisfy the ECE design requirement. Your project adviser will determine whether or not your particular project satisfies this requirement. To have your project satisfy this requirement, you must first obtain approval from your adviser, based on a written project plan submitted with your sign-up form prior to beginning your work. Your plan must include appropriate specifications and a description of constraints. Final approval of satisfying the design requirement will be signified by your adviser checking the appropriate box on your final report grade form.

Engineering Standards

The thesis projects are also designed to prepare students for engineering practice. Therefore, it is required that the project incorporates appropriate engineering standards, which should be clearly identified in the thesis report. For clarity we ask all Seniors to provide a focused section in the Appendix of the thesis report that will list the engineering standards and broadly accepted industrial conventions used in the thesis with a short description of their role/use in the thesis.

For your reference, a list of engineering/industrial standard examples (by no means exhaustive) and an example Appendix is provided here. If you have trouble identifying any standardized software, hardware, tools, framework used in your project, please consult with your Thesis Adviser.

Human and Animal Subjects Research Standards

Research involving human subjects and/or animal subjects is regulated, and each student will be asked to attest that their project does or does not include human subjects and does or does not include animal subjects. Students whose projects include either human or animal subjects must obtain approval and authorization before start of work on those parts of the project. Human and/or animal subjects research that was performed without prior approval and authorization may not be submitted, and violation of this policy may result in a grade of INC or F.

For human subjects, this will go through Princeton's IRB (Information available here). For animal subjects, this will go through Princeton's IUCAC (Information available here). For students who are

unsure of whether or not their research counts as regulated human subjects and/or animal subjects research, a short project description can be submitted to either governing body for an official determination. Students whose projects involve human and/or animal subjects but who are working with a faculty adviser are encouraged to discuss with their adviser, since often a student's project approval and authorization can be part of the faculty's approval and authorization.

P/D/F, Breadth, Concentration and Departmental Course

Independent work cannot be taken P/D/F. It cannot be used to fulfill the ECE breadth or concentration requirements, but it can be counted as a Department course.

COURSE LEARNING OBJECTIVES

The following learning objectives (student outcomes) constitute the minimum skills that every student must acquire through the senior thesis/independent work experience. These objectives will be used, in part, to evaluate the student's work and in the assignment of a grade.

Course Learning Objectives	**ABET Criterion 3**
An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science and mathematics	1
An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental and economic factors	2
An ability to communicate effectively with a range of audiences	3
An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental and societal contexts	4
An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	5
An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	6
An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	7

The Senior Thesis in the Department of Electrical and Computer Engineering is also designed to provide students with a major design experience based on the knowledge and skills acquired in earlier course work and incorporating appropriate engineering standards and multiple realistic constraints.

THE GRADED ELEMENTS OF SOPHOMORE/JUNIOR INDEPENDENT WORK

During the semester you need to complete the following three elements of Independent Work.https://ece.princeton.edu/

1. **Project Proposal Form (ungraded but required)** The aim of the project proposal is to ensure that you are headed in the right direction. The majority of problems that arise in independent work are due to a late start. We want to see that you have an adviser selected, a project with a well-defined objective, and some idea of the steps it will take to achieve your objectives. Obviously, projects evolve as the semester progresses; you are not bound by your original proposal, but it will help guide your research. **Due by online form submission** (for issues contact ece-ug@princeton.edu **by Wednesday, February 11, 2026.** **Project Proposal Form**

   If you are having trouble finding an adviser, look at https://ece.princeton.edu/academics/undergraduate/student-projects/independent-project-ideas

   For a list of ideas. You are welcome to work on a project outside of a faculty research area, but will need to find a faculty mentor for these projects. If you are having difficulty, contact either the Undergraduate Program Coordinator (Lori Bailey), IW/Senior Thesis coordinator, (Jean Bausmith) faculty instructor for the Independent Work ECE courses (Paul Prucnal) or Directors of Undergraduate Studies (Prateek Mittal, Alejandro Rodriguez, Hossein Valavi).

   It is also recommended to read the section entitled, "**GENERAL ADVICE REGARDING FORMAT OF FINAL REPORT**" (below) as it contains some questions that are thought provoking and helpful when creating a project proposal.

2. **Poster Presentation (10% of grade) – Wednesday, April 29, 2026 at 12:00 in the E-Quad, 3rd floor, B and C wings.** All independent work students, a majority of the faculty, and many interested students throughout the department will be in attendance. A group of faculty, including your adviser, will provide a grade for your poster presentation. You should prepare a poster that can clearly express your work this semester, including project goals, background and context, accomplishments and results, methodology, conclusions and future directions. Remember, posters need to be printed, so plan ahead!

3. **Final Written Report (90% of grade**) Students doing independent work are required to provide via Form submission a copy of the Independent Work report to the Departmental Undergraduate Office by 11:59pm on **Tuesday, April 28, 2026**. Submission Form

In addition, by the same deadline, they should email a pdf file of the report to **their adviser and second reader (if applicable)**.

Senior Thesis

A senior thesis is mandatory for all ECE majors.

A spring poster session (10% of grade) will be held on **Monday, May 4, 2026 at 1:30 pm in the Friend Center Convocation Room 113 and upper atrium,** to showcase your work to date. The final report and poster session will represent your grade for ECE499.

The written thesis (90% of grade) is the document submitted as the ECE 499 project report. The thesis is formatted specifically to the stated University regulation that will be provided to the student. Thesis students are required to email a pdf of the thesis to your adviser and second reader (if applicable) and the **UG office by submitting to:** Submission Form The senior thesis due date is **Monday, April 13, 2026**.

If no extension is formally granted and you are late in handing in your final report your grade will be adjusted by -⅓ letter grade. For example, a report that is otherwise graded as an A will be reduced to an A-. Reports that are more than one week late will suffer an additional -⅓ letter grade penalty.

Poster Presentation (10% of grade) – at a poster session on **Monday, May 4, 2026, at 1:30pm in the Friend Center Convocation Room 113,** during Reading Period. A group of faculty will provide a grade for your poster presentation. You should prepare a poster that can clearly express your work this semester, including project goals, background and context, accomplishments and results, methodology, conclusions and future directions.

Senior Thesis TA Check-Ins

In order to keep students on track, each senior will sign up to have check-ins every two weeks (excluding fall break) with TAs. Sophomores and juniors will not check-in with TAs. These meetings can be very quick for students who are on track just to say some progress has been made, or can be up to 15 minutes for students who have questions or need guidance. Students who need additional help can schedule additional time with TAs as needed as well, but this does not replace the requirement to primarily meet with your advisor. Sign-ups will be via canvas. In order to keep students on track, each **senior** will need to check-in to two rather than four for the spring. Feb 2 - Feb 13, Feb 16 - 27, Mar 2 - 20 (Inc Spring Break), Mar 23 - Apr 3. **TA check-ins cannot be rescheduled! Failure to check-in with TAs regularly will result in -1/3 letter grade for missing more than 3/6 TA check-ins without advance notice.**

Senior Thesis Guidelines

The written thesis is the document submitted as the ECE 499 project final report. The thesis is to be submitted by spring semester deadline per University regulations as previously stated. The thesis is to be formatted specifically to the same format as a two semester final report.

The adviser and a second reader (second reader is only required if the adviser is outside of ECE. The second reader is then required to be an ECE faculty member) student, will examine the thesis. Each member of the committee will provide both a quantitative and a written evaluation of the thesis, which must include a written statement recommending the project for recognition as a senior thesis.

GENERAL ADVICE REGARDING FORMAT OF FINAL REPORT

The most frequently asked questions concern what is actually in the final report (how much detail, how long should it be, etc.). This section of the guide is a strictly informal set of guidelines describing the basic components of a final report. Since projects differ widely, it is impossible to develop a general set of guidelines that applies equally to all students. Regardless of how well you think your own research fits the following guidelines, you should talk to your adviser to determine the most appropriate style of presentation for your own work.

The essence of any scholarly work is to establish the following:

- Definition of the problem and review of the literature
- Presentation of your particular contribution
- Identification of fruitful areas of further research that others in the field may use to guide their own work

Toward these three goals, the following list of questions may prove useful for organizing both your research effort and the final writing of the final report.

What are you looking at?

You must begin by defining your problem. In the introduction of the final report, you want to do this in a general way that gives the reader a sense of the scope of the project and a basic understanding of your problem. For example, you may be solving a problem of interest to a particular company, or developing a new approach to a problem that is of interest to the research community as well.

Why are you looking at it?

Motivate your work. Explain to the reader why the problem is interesting and important. Establish who will benefit from your work and why. Note that you do not have to get the whole world

interested in your work. Most independent work involves the application of existing techniques to particular problems, and hence the people who will be most directly interested may be those working for a particular company, engineers working on a specific project, and so on.

WHO ELSE LOOKED AT IT?

Now that we (the readers) have a rough idea of what your problem is, you must firmly establish what the state of the art is in the area. This is particularly critical if you wish to claim that you have a better way of solving/approaching a problem than has appeared previously in the literature. List others who have worked on the same or similar problems and briefly describe their work. If you are duplicating the approach used by others in the field, but using a different data set that is of particular interest to you, say so and describe why you have chosen this approach over others that may have been used.

HOW ARE YOU LOOKING AT IT?

At this point, you may need to review your problem again but at a much higher level of detail, introducing any mathematical notation required and describing any subtle aspects of your problems that may in fact be the central component of your research but which were too detailed to put in the introduction. Empirical research, which involves gathering data to prove some relationship, can often be introduced by citing one or more hypotheses which you feel your research will prove (or disprove).

Not all work is conducive to initial statements of hypotheses, particularly methodological topics which are aimed at better solutions to existing problems (the implicit hypothesis is that your method is better than others, but this need not be stated as such). In any event, unless your work is purely theoretical, you should describe in detail your experimental design: how you structured your data collection, problems you encountered, and how you conducted your experiments. The description should be sufficiently detailed to allow another researcher to duplicate your efforts. A key part of your description should be a clear list of major assumptions you are making and why you are making them. It is useful at the same time to indicate which assumptions are perfectly reasonable (e.g. other researchers have used the same assumption and have obtained good results) and which are likely to affect your results but are required for time/budget reasons.

WHAT ARE THE LIMITATIONS OF YOUR WORK?

One of the most difficult aspects of research is understanding exactly what you did and what you did not do. If you were limited by your data, explain how you think this might affect the generality of your conclusions. Discuss openly any shortcuts required due to time/budget/data availability constraints. Do not try to claim credit by stating that you feel that your method will work in more general situations if you have done only limited testing. At the same time, do not feel you are getting off the hook by over-qualifying your work (e.g. "Because of such and such restrictions, no valid conclusions can be drawn until more extensive experiments are carried out using so and so's

data or equipment."). Clearly drawing the line between what you did and did not do is a central step in the scientific method since it helps define the state of the art.

WHAT ARE YOUR CONCLUSIONS?

In view of the limitations above, what conclusions can you draw from your research? Because your conclusions are often inextricably intertwined with the limitations of your research, both questions are often answered simultaneously. It may be useful to discuss limitations of specific aspects of your work while you are describing the work itself, but defer a discussion of how such limitations actually impact your results until later. Your section on conclusions is usually brief, and should specifically and clearly describe your contributions to the field. Frequently, researchers familiar with the field will start by reading your conclusions and, depending on your claims and then decide whether to read the report itself. Again, do not underrate your work, but do not claim to have solved problems that are not firmly substantiated in the body of the final report.

WHAT NEXT?

Areas for future research. Now that you are an expert in your particular area, you should have both a narrow understanding of a well-defined problem as well as a broader understanding of the field as a whole. As such an authority, it is now your responsibility to guide others in the fields that do not have the benefit of your particular experience in directions that you feel will provide the greatest good. Such recommendations are usually based on an evaluation of the major weaknesses in your own work, in which case you might recommend how others (preferably with more time and money than you enjoyed) could overcome these weaknesses. Be sure, however, to specify those weaknesses that you feel would have the greatest impact on your conclusions. Some assumptions that you may have made may be perfectly reasonable, in which case a more accurate model would not improve the final results.

FORMAT OF THE FINAL REPORT

The Independent Work report should look like a professional document – 12-point font (Times New Roman preferred), appropriate margins, double or 1½ - spaced.

The report should be divided into chapters, numbered sequentially, and a table of contents provided immediately after the first 4 pages (i.e., starting page 5).

Supplemental material (e.g., lengthy computer code used in the Independent Work, or long tables of supporting data) should be provided in appendices after the concluding chapter, and numbered sequentially.

The report should contain a proper bibliography. References should be numbered in order they appear in the report (preferred); alternatively, they should be referenced as [author, year]. The

complete reference list should be provided at the end, either sequentially by number, or alphabetically by author, depending on which of the above referencing schemes is used. The engineering standards used in the thesis and their role should be clearly identified in the written thesis. This may be done through footnotes, references, or a focused section of the thesis report.

All non-original text should be properly attributed. Failure to cite sources for ideas, tables, text or diagrams is a violation of Princeton University's code of ethics. If you are unsure about how to cite sources properly, ask your adviser.

Charts, tables, diagrams, etc., should be accompanied by proper captions, and should be appropriately referenced in the text. Footnotes, if necessary, should be used sparingly. If the numbered referencing scheme is followed, references could be used in lieu of footnotes.

The final report should be printed on one side of standard-size paper (8 ½ X 11 inches). The typing should be "double-spaced" or 1.5 spaced. The latter is recommended, with the exception of footnotes and bibliography, which should be single-spaced. All margins should be 1 inch or larger. All pages must be numbered. The format of the first four pages should be as given below.

INSTRUCTIONS FOR FIRST FOUR PAGES

Page 1 – The first page should include the title of the project, name of author, date, name of adviser(s), and the statement:

Submitted in partial fulfillment
of the requirements for the degree of
Bachelor of Science in Engineering
Department of Electrical and Computer Engineering
Princeton University

Page 2 – The second page should contain the following statements. If you answer that your project contains regulated human and/or animal subjects research, any and all approval and authorizations must be included in the appendices:

I hereby declare that this Independent Work report represents my own work in accordance with University regulations.

I hereby declare that this Independent Work report (does / does not) include regulated human subjects research.

I hereby declare that this Independent Work report (does / does not) include regulated animal subjects research.

Your Signature

Your Name

Page 3 – The third page should contain the title of the Independent Work project, your name, and an abstract all of which fit within ONE page:

Title

Author

Abstract - text

Page 4 – The fourth page, entitled "ACKNOWLEDGMENTS", should contain relevant acknowledgments, citing your debt to individuals who contributed to your thesis, as well as any sources of financial support (departmental/SEAS funds, federal grants – check with your adviser for specific details). Any dedications should be part of this page.

TIPS ON ORGANIZING YOUR TIME

One of the most common mistakes made by students doing independent work is underestimating how much time it takes to complete certain tasks, in particular the actual writing of the final report. Avoid undertaking overly ambitious projects, because they often take much longer than you expect. Order any parts you need as early as possible.

It may be helpful to divide the effort into three primary tasks:

1. Defining the problem and reviewing the literature through survey reading and digesting the background material. Plan the project, week by week, so that you have a finished project well before the deadline.

2. Doing the work (including experimental work, if applicable). Don't hesitate to redirect, redefine or even change your project if you are stuck. One objective of independent work is to recognize when it is appropriate to quit and try something else.

3. Writing the final report. Make an outline for the final write up well in advance of the writing so no time is wasted reorganizing or rewriting major sections in the last minutes before the deadline.

If your research is fairly well defined (usually with the help of your adviser), then the first stage may be somewhat reduced. Naturally, the three tasks will overlap, since you may have to do additional literature review when you finally settle on a specific problem, and it is often useful to begin writing certain sections of the final report while the actual research is in progress.

Please do not underestimate how long it takes to write a good final report/thesis. START EARLY!

Extensions

Extensions are not normally given. If, for any reason you must be away from campus when one of these reports is due, plan to turn the report in early. Extensions for turning in the IW report/thesis will be granted only in the case of illness (or family emergencies) and only when such illness or emergency makes it impossible to complete the final report on time. All final report extensions must be requested in writing and turned in to the Director of Undergraduate Studies prior to the deadline for the final report. Each request must be approved first by the student's adviser and then by the Director of Undergraduate Studies and finally by the Office of the Dean of the College. Extensions will not be granted for unexpected delays due to problems in your experiment, analysis, simulations or technical issues related to writing your thesis.

Additional Resources

Writing Support

Located at 2 New South, the Writing Center offers free one-on-one conferences with experienced fellow writers trained to consult on assignments in any discipline. Special 80-minute conferences are available for JP and Senior Thesis writers, who may sign up to work with a graduate student fellow from the department of their choice. The Writing Center also holds 50 minute regular conferences seven days a week, and drop-in hours Sunday through Thursday evenings. View their website to make appointments at https://writing.princeton.edu

Thesis Archive

The Princeton Thesis Archive is a great resource to see what senior theses have been completed in the past in the department and can enable finding potential good advisers. In addition the following website provides details on how to archive your thesis and how to handle issues around embargoing

release of thesis for a fixed period of time and for a compelling reason (e.g. patent being filed, publication being submitted to venue with strict disclosure requirements, or privacy concerns): https://undergraduateresearch.princeton.edu/independent-work/thesis-archive

GRADING

The following are examples of the grading sheets that will be used to evaluate the graded elements of independent work.

EXAMPLE OF AN ADVISER GRADING SHEET FOR THE FINAL REPORT/THESIS

Name of Student:

Title of Independent Work:

Please provide scores, write your descriptive evaluation, and overall grade based on the following criteria:

1. **Originality and Creativity** – e.g., clear independence and novel thinking; going beyond literature and adviser.

1 (low)	2	3	4	5 (high)

2. **Technical Quality** – Consistency, control, reliability/reproducibility.

- Quality of the experiment/device/system/process design

1 (low)	2	3	4	5 (high)

- Quality of data and data analysis

1 (low)	2	3	4	5 (high)

3. **Communication Quality**
   a. **Completeness** – Background, motivation, completion and accomplishments of project, and future work. Indicate if publishable; how much follow-up work is needed?
   b. **Level of Scholarship** -- literature search and reference.
   c. **Writing Quality** – Crisp, well written, needs little editing or revision.

1 (low)	2	3	4	5 (high)

4. **Relative Standing** – How does the report compare with Independent Work reports you have seen in the past five years?

Below Average	Average	Top 30%	Top 5%

DESCRIPTIVE EVALUATION

Guide:
(A+) - Truly exceptional
(A) - Very Good to Excellent; well above nominal expectations
(B) - Good to very good – covers everything that is expected
(C) - Adequate to fair; would have liked to see more
(D) - Marginal
(F) - Unacceptable

Example of Oral Presentation Grading Sheet for Advisers and Faculty Committee:

Name of Student:

Title of Presentation:

Please provide a short description of your evaluation, along with a grade (A-F).

In making your descriptive evaluation, you may wish to consider the following, given the time allotted for the presentation (10 minutes):

1. Did the student provide adequate background for the technical presentation?
2. Was the motivation for the project well presented?
3. Did the presentation give a clear idea of the methodology and technical merit of the work?
4. Did the student present the accomplishments clearly?
5. Were the conclusions justified and well articulated?
6. Did the student suggest follow-up investigations or directions as a result of the work done?
7. How did the work compare with that of your other students in this course in the past 5 years – best, one of the better ones, average, below average, one of the lower achievers?

Descriptive Evaluation

Overall Grade:

Guide:
(A+) – Truly exceptional
(A) – Very good to excellent; well above nominal expectations
(B) – Good to very good; covers everything that is expected
(C) – Adequate to fair; would have liked to see more
(D) – Marginal
(F) – Unacceptable